1. Two converging lenses with focal lengths of 40 cm and 20 cm are 10 cm apart. A 2.0-cm-tall object is 15 cm in front of the 40-cm-focal-length lens

- **A.** Use ray tracing to find the position of the image. Do this accurately using a ruler or graph paper with a grid, then make measurements on your diagram.
- **B.** Calculate the image position and height. Compare with your ray tracing answers in **A.**
- **C.** Determine the system magnification

2. A converging lens with a focal length of 40 cm and a diverging lens with a focal length of -40 cm are 160 cm apart. A 3.0-cm-tall object is 60 cm in front of the converging lens.

- **A.** Use ray tracing to find the position of the image. Do this accurately using a ruler or graph paper with a grid, then make measurements on your diagram.
- **B.** Calculate the image position and height. Compare with your ray tracing answers in **A.**
- **C.** Determine the system magnification

3. A 2.0-m-tall man is 10 m in front of a camera with a 15-mm-focal-length lens. How tall is his image on the detector?

4. You have been asked to build a telescope from a 2.0× magnifying lens and a 5.0× magnifying lens.

- **A.** What is the maximum magnification you can achieve?
- **B.** Which lens should be used as the objective?
- **C.** What will the length of your telescope be?

5. A 1.0-cm-object sits 5-cm in front of a converging lens with a focal length of 10 cm, which in turn is 5 cm in front of a **convex mirror** with a focal length of -30 cm. What are the position, height, and orientation of the final image? Give the position as a distance to the right or left of the lens. (Hint: the light bounces off the mirror and goes back through the lens, so the lens gets used a second time.)

6. An object of 6-cm is viewed through a magnifying glass of focal length f=10 cm. How large does it appear?

7. Open the **Build a telescope** widget in §3 of the Day 1.7 notes. Pick an f_o and an f_e, then slide the eyepiece until the rays leave parallel. What do you notice about the separation of the lenses when you've found it? Compare the ratio θ_e/θ_o the widget measures with $-f_o/f_e$.